

ECE 4644 / 5610 Satellite Communications

Homework Assignment #2 - SOLUTIONS

Problem #1

According to a NASA source, the Artemis II mission will be boosted from an initial LEO orbit to a High Earth Orbit (HEO) before launching towards the moon. The lowest altitude for this orbit will be 185 km and the highest altitude will be 73,600 km.

- Determine the semi-major axis, semi-minor axis, period, and eccentricity of the orbit.
- Sketch the orbit approximately to scale and include the Earth centered properly. For perspective, trace out a GEO orbit.
- Indicate the satellite positions on the sketch where the true anomaly, ϕ_o , is 0° , 90° , 180° , and 270° . Label the perigee and apogee.
- For each of these four positions determine the remaining orbital plane coordinate r_o and (x_o, y_o) . Arrange your results in a Table.
- Previously you solved for the ratio of the satellite velocities at apogee and perigee for the initial LEO orbit. Now apply Kepler's second law again (i.e., do not use a canned equation) to determine the actual velocities at apogee and perigee for the HEO orbit. (Hint: consider that the average rate at which area is swept out, $\langle dA/dt \rangle$, can be found by considering the area of the ellipse and the period of the orbit.)

Problem #2

A certain satellite is described in the NASA prediction bulletin as having an orbit with a semimajor axis of 7194.9 km and an eccentricity of 9.5981×10^{-3} . The satellite performs 14.2171404 revolutions per day. At time $t = 0$ the mean anomaly is 204.9779° . (This is the *epoch* of the orbital elements given in the bulletin, i.e., the moment when the 'snapshot' of the satellite in its orbit is taken and the orbital data recorded.)

- Find the orbital period, average angular velocity, and the time of perigee relative to $t = 0$.
- Using *Kepler's equation* ($M = E - e \sin E$) find the eccentric anomaly at $t = 0$. (Note that the standalone values of M and E in this equation must be expressed in radians.)
- Finally, determine the radial distance from the center of Earth and the true anomaly at time $t = 0$. A sketch will help you resolve any ambiguities.

Problem #3

- In your own words explain the terms *right ascension* (RA) and *declination* (δ).
- What are the (RA, δ) coordinates of the Sun on the first days of Spring, Summer, Winter, and Fall?

- c. Explain whether the (RA, δ) coordinates of a satellite in GEO orbit are changing or staying the same (assume a Keplerian orbit, i.e., no perturbing forces).
- d. A satellite in *polar orbit* passes over the North and South poles of Earth. Describe how its (RA, δ) coordinates vary over an orbit (assume a Keplerian orbit and consider that the orbit is fixed in orientation relative to the distant stars).

Problem #4

6/ With the help of careful sketch explain the meaning of the angles *inclination*, *right ascension of the ascending node*, and *argument of perigee* in reference to the orbit of a satellite. List the units for each angle and its range (from minimum value to maximum value).

Note that inclination can exceed 90° resulting in *retrograde* motion, that is, the satellite moves in the direction opposite to Earth's rotation. A satellite in a *prograde* orbit moves in the direction of Earth's rotation (which is more usual). Include both a prograde and a retrograde orbit in your sketch.

Problem #5

8/ The first commercial communications satellite was Intelsat 1, also known as Early Bird. It was launched in August 1965 into a geostationary orbit positioned at 28.0° W longitude. The aim was to provide trans-Atlantic service. We will examine the scope for making connections with a satellite at this location.

Consider these potential locations for siting Earth stations: Chicago (41.8° N, 87.6° W), Santiago, Chile (33.45° S, 70.67° W), Paris, France (48.85° N, 2.35° E), and Johannesburg, South Africa (26.2° S, 28.0° E).

- a. Calculate the look angles for each station.
- b. Comment on the suitability of each station for communications with the satellite in the C, Ku, and Ka bands.
- c. Noting the geographic distribution of these cities, suggest alternatives for any that could not support communications in all three bands. Demonstrate suitability by calculation.

$$\Sigma = 40$$

ECE 5610: Satellite Communications

Homework #2

Problem #1

Constants and Initial Values

- Mean Earth radius: $R_E = 6378.137$ km
- Earth gravitational parameter: $\mu = 3.9860044 \times 10^5 \text{ km}^3/\text{s}^2$
- Perigee altitude: $h_p = 185$ km
- Apogee altitude: $h_a = 73600$ km

(a) Orbital Parameters

First compute the orbital radii (measured from Earth center):

$$\begin{aligned}r_p &= R_E + h_p = 6378.137 + 185 = 6563.137 \text{ km}, \\r_a &= R_E + h_a = 6378.137 + 73600 = 79978.137 \text{ km}.\end{aligned}$$

Semi-major axis (a):

$$a = \frac{r_p + r_a}{2} = \frac{6563.137 + 79978.137}{2} = 43270.637 \text{ km.}$$

Eccentricity (e): (equivalently $r_p = a(1 - e)$ and $r_a = a(1 + e)$)

$$e = \frac{r_a - r_p}{r_a + r_p} = \frac{79978.137 - 6563.137}{79978.137 + 6563.137} = 0.8483235.$$

Semi-minor axis (b):

$$b = a\sqrt{1 - e^2} = 43270.637\sqrt{1 - (0.8483235)^2} = 22910.859 \text{ km.}$$

Orbital period (T): (Kepler's 3rd law)

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} = 2\pi\sqrt{\frac{(43270.637)^3}{3.9860044 \times 10^5}} = 89577.908 \text{ s} \approx 24 \text{ h } 52 \text{ min } 57.9 \text{ s.}$$

(b) & (c) Orbit Sketch (approximately to scale)

For the sketch, Earth is placed at a focus. The focus offset from the ellipse center is

$$c = ae = (43270.637)(0.8483235) = 36700.0 \text{ km.}$$

In the plot below, we use **1 unit = 1000 km**.

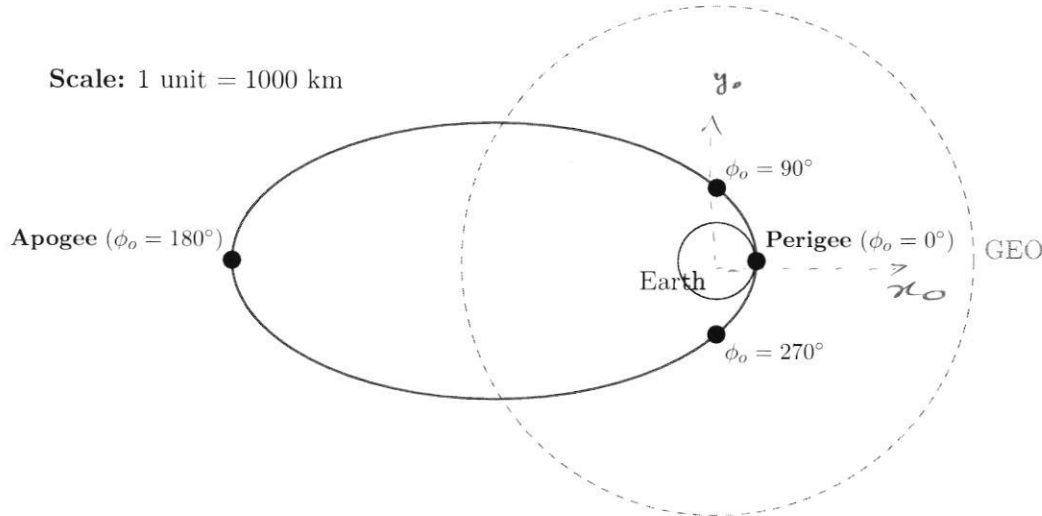


Figure 1: HEO ellipse (Earth at a focus) with GEO reference circle.

(d) Orbital Plane Coordinates Table

Using the polar orbit equation (focus at Earth):

$$r_o(\phi_o) = \frac{a(1 - e^2)}{1 + e \cos \phi_o}, \quad x_o = r_o \cos \phi_o, \quad y_o = r_o \sin \phi_o.$$

First compute the semi-latus rectum:

$$p = a(1 - e^2) = 12130.801 \text{ km.}$$

Position	ϕ_o (deg)	r_o (km)	x_o (km)	y_o (km)
Perigee	0°	6563.137	6563.137	0
Quadrature	90°	12130.801	0	12130.801
Apogee	180°	79978.137	-79978.137	0
Quadrature	270°	12130.801	0	-12130.801

Table 1: Orbital-plane coordinates at $\phi_o = 0^\circ, 90^\circ, 180^\circ, 270^\circ$.

(e) Velocities using Kepler's Second Law

Idea: Kepler's Second Law states the areal rate is constant:

$$\frac{dA}{dt} = \text{constant}.$$

At perigee and apogee the velocity is perpendicular to the radius vector ($\vec{v} \perp \vec{r}$), so the swept area over a small time dt is approximately a triangle of base r and height $v dt$:

$$dA = \frac{1}{2}r(v dt) \Rightarrow \frac{dA}{dt} = \frac{1}{2}rv.$$

Therefore, at perigee and apogee:

$$\frac{1}{2}r_p v_p = \frac{1}{2}r_a v_a \Rightarrow r_p v_p = r_a v_a \quad (\text{from HW\#1}).$$

To obtain the *actual* values of v_p and v_a , we need the numerical value of $\frac{dA}{dt}$.

Compute the areal rate dA/dt : Over one full orbital period T , the radius vector sweeps out the entire ellipse area,

$$A_{\text{ellipse}} = \pi ab.$$

Since $\frac{dA}{dt}$ is constant in a Keplerian orbit, the instantaneous areal rate equals the average areal rate:

$$\frac{dA}{dt} = \frac{A_{\text{ellipse}}}{T} = \frac{\pi ab}{T}.$$

Numerically,

$$\frac{dA}{dt} = \frac{\pi(43270.637)(22910.859)}{89577.908} = 34768.313 \text{ km}^2/\text{s}.$$

Velocities at perigee and apogee: Using $\frac{dA}{dt} = \frac{1}{2}rv$ at those points,

$$v = \frac{2}{r} \frac{dA}{dt}.$$

Hence

$$v_p = \frac{2}{r_p} \frac{dA}{dt} = \frac{2(34768.313)}{6563.137} = 10.5950 \text{ km/s},$$
$$v_a = \frac{2}{r_a} \frac{dA}{dt} = \frac{2(34768.313)}{79978.137} = 0.86945 \text{ km/s}.$$

1 Problem #2

Given Parameters and Constants

- Semi-major axis: $a = 7194.9$ km
- Eccentricity: $e = 9.5981 \times 10^{-3}$
- Mean anomaly at the epoch: $M_0 = 204.9779^\circ$ at $t_0 = 0$
- Gravitational parameter: $\mu = 3.9860044 \times 10^5 \text{ km}^3/\text{s}^2$

(a) Period, Average Angular Velocity, and Time of Perigee

Orbital period T :

$$T = 2\pi \sqrt{\frac{a^3}{\mu}} = 2\pi \sqrt{\frac{(7194.9)^3}{3.9860044 \times 10^5}} = 6073.63 \text{ s } (\approx 101.23 \text{ min}).$$

Average angular velocity, η :

$$\eta = \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}} = 1.034503 \times 10^{-3} \text{ rad/s.}$$

$\Rightarrow$ mn: sec format
101^{mn} 13.8^{sec}

Time of perigee passage relative to $t_0 = 0$:

Mean anomaly evolves linearly as

$$M(t) = \eta(t - t_p),$$

where t_p is the perigee passage time. At the epoch $t_0 = 0$,

$$M_0 = \eta(0 - t_p) = -\eta t_p \Rightarrow t_p = -\frac{M_0}{\eta}.$$

Convert M_0 to radians:

$$M_0 = 204.9779^\circ \left(\frac{\pi}{180} \right) = 3.577539 \text{ rad},$$

so

$$t_p = -\frac{3.577539}{1.034503 \times 10^{-3}} = -3458.22 \text{ s} \approx -57.64 \text{ min.}$$

Interpretation: perigee occurred about 57.6 minutes *before* the epoch $t = 0$.

(b) Eccentric Anomaly at the Epoch $t = 0$

Goal: Find the eccentric anomaly E_0 at the epoch given M_0 and e .

Why iteration is needed: Kepler's equation is

$$M_0 = E_0 - e \sin E_0,$$

which is transcendental (since E_0 appears both inside and outside $\sin(\cdot)$), so we solve for E_0 numerically.

Initial guess: Because $e \ll 1$ (nearly circular orbit), the correction term $e \sin E_0$ is small and thus $E_0 \approx M_0$. We therefore use the standard initial guess

$$E_0^{(0)} = M_0.$$

Convert M_0 to radians:

$$M_0 = 204.9779^\circ \left(\frac{\pi}{180} \right) = 3.577539 \text{ rad.}$$

Newton's method: Define

$$f(E) = E - e \sin E - M_0, \quad f'(E) = 1 - e \cos E.$$

Newton iteration:

$$E_0^{(k+1)} = E_0^{(k)} - \frac{f(E_0^{(k)})}{f'(E_0^{(k)})} = E_0^{(k)} - \frac{E_0^{(k)} - e \sin(E_0^{(k)}) - M_0}{1 - e \cos(E_0^{(k)})}.$$

Iterations (stop when $|E_0^{(k+1)} - E_0^{(k)}| < 10^{-8}$ rad):

k	$E_0^{(k)}$ (rad)	$ E_0^{(k)} - E_0^{(k-1)} $ (rad)
0	3.577539000	—
1	3.573521230	4.018×10^{-3}
2	3.573521263	3.3×10^{-8}

Therefore,

$$E_0 \approx 3.573521263 \text{ rad} = 204.7477^\circ.$$

(c) Radial distance r_0 and true anomaly ϕ_0 at $t = 0$

Using the eccentric anomaly at the epoch from part (b), E_0 , we compute:

1) Radial distance r_0 . The radius in an elliptic orbit in terms of eccentric anomaly is

$$r_0 = a(1 - e \cos E_0).$$

Substitute $a = 7194.9$ km, $e = 9.5981 \times 10^{-3}$, and $E_0 = 3.573521263$ rad:

$$\begin{aligned} \cos E_0 &= \cos(3.573521263) = -0.9081601, \\ 1 - e \cos E_0 &= 1 - (9.5981 \times 10^{-3})(-0.9081601) = 1.0087166, \\ r_0 &= (7194.9)(1.0087166) = \mathbf{7257.62 \text{ km.}} \end{aligned}$$

2) True anomaly ϕ_0 (two-solution ambiguity and selection). From E_0 , compute

$$\cos \phi_0 = \frac{\cos E_0 - e}{1 - e \cos E_0}.$$

Using the values above:

$$\cos \phi_0 = \frac{-0.9081601 - 0.0095981}{1.0087166} = -0.9098276.$$

Because the inverse cosine returns two angles in $[0^\circ, 360^\circ)$,

$$\phi_{0,1} = \arccos(-0.9098276) = 155.48^\circ, \quad \phi_{0,2} = 360^\circ - \arccos(-0.9098276) = 204.52^\circ.$$

Conceptual choice: At the epoch, the mean anomaly is $M_0 = 204.98^\circ > 180^\circ$, so the satellite is *past apogee* and lies in the **third quadrant** of the orbital plane ($x_0 < 0$, $y_0 < 0$), hence it must have $\sin \phi_0 < 0$ and $\phi_0 > 180^\circ$. Therefore, we select the Quadrant-III solution:

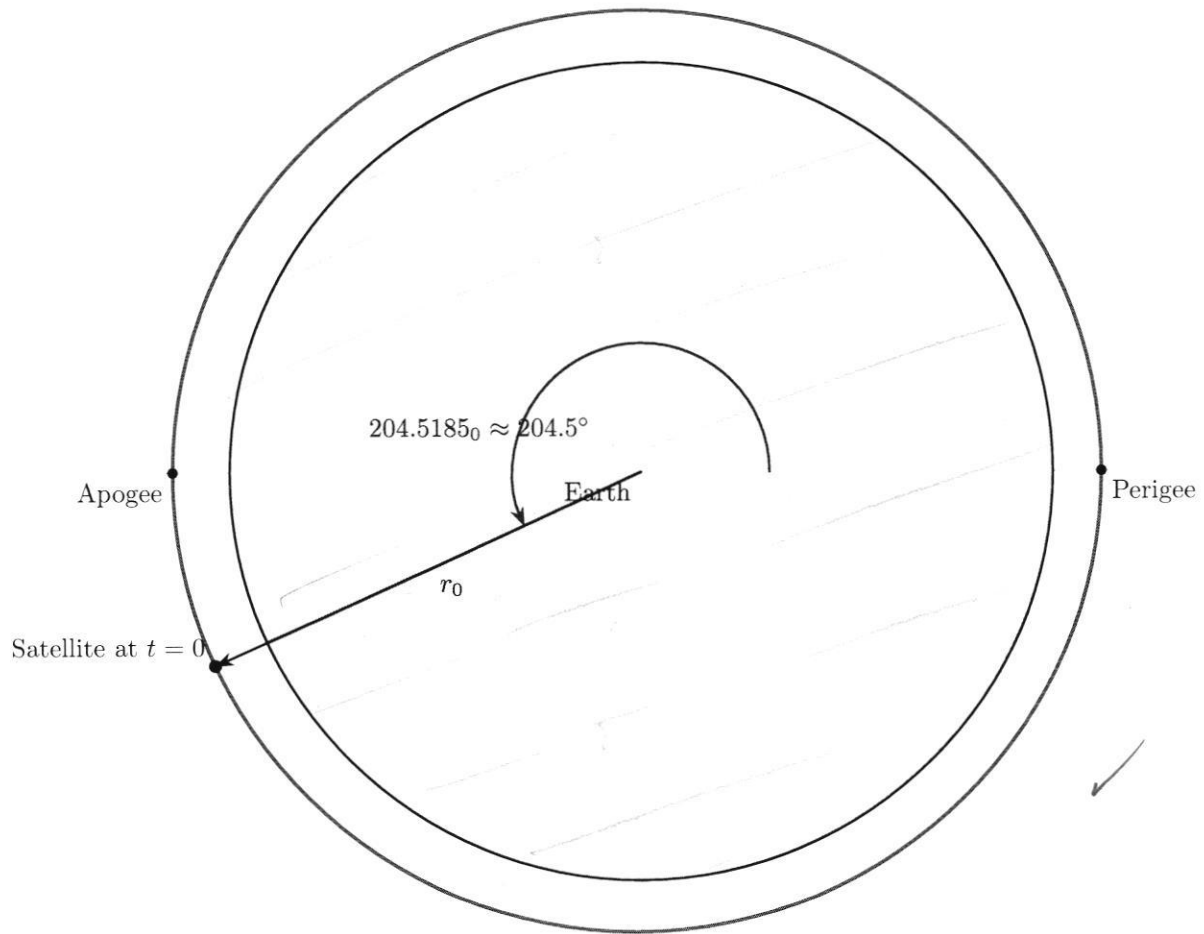
$$\phi_0 = \phi_{0,2} = 204.52^\circ = 3.56952 \text{ rad.}$$

Final results at $t = 0$:

$$r_0 = 7257.62 \text{ km},$$

$$\phi_0 = 204.52^\circ (3.56952 \text{ rad}).$$

Orbital Visualization (at the Epoch $t = 0$)



Problem #3

- (a) In your own words explain right ascension (RA) and declination (δ).

Considering the sky as a *celestial sphere* with Earth at the center. If we project Earth's equator onto the sky, we get the **celestial equator**. Then RA and declination are simply the *longitude and latitude on that sky-sphere*.

- **Declination δ (sky-latitude):** how far the object is above (+) or below (−) the celestial equator. It is measured along the object's hour circle, with range $-90^\circ \leq \delta \leq +90^\circ$.
- **Right ascension α (sky-longitude):** how far *eastward* the object is along the celestial equator. To measure a longitude, we need a zero reference direction; for RA the zero direction is the **vernal equinox direction** (First Point of Aries). RA is measured eastward from that direction, with range $0 \leq \alpha < 360^\circ$ (often written as 0 to 24 hours, where $24 \text{ h} = 360^\circ$).

- (b) What are the (RA, δ) coordinates of the Sun on the first days of Spring, Summer, Winter, and Fall?

Let Earth's obliquity be $\varepsilon \approx 23.44^\circ$. On the equinoxes the Sun lies on the celestial equator ($\delta = 0^\circ$). At the solstices it reaches its maximum north/south declination ($\pm\varepsilon$), and its RA advances by about 6 hours per season.

Season start	Sun RA α	Sun declination δ
Spring (vernal equinox)	0 h (0°)	0°
Summer (summer solstice)	6 h (90°)	$+\varepsilon \approx +23.44^\circ$
Fall (autumnal equinox)	12 h (180°)	0°
Winter (winter solstice)	18 h (270°)	$-\varepsilon \approx -23.44^\circ$

- (c) Explain whether the (RA, δ) coordinates of a satellite in GEO orbit are changing or staying the same (Keplerian).

For an ideal GEO (circular and equatorial), the satellite remains over a fixed *Earth-fixed* longitude, so it looks "stationary" to an observer on Earth. However, RA/declination are *inertial* sky-coordinates (referenced to the stars). In an inertial frame the GEO satellite still completes one revolution per sidereal day.

Therefore:

$$\delta(t) \approx 0^\circ \quad (\text{stays on the celestial equator}),$$

$$\alpha(t) \text{ increases steadily (wraps through } 360^\circ \text{ per sidereal day).}$$

- (d) A satellite in polar orbit passes over the North and South poles. Describe how its (RA, δ) vary (Keplerian, orbit fixed to stars).

For a Keplerian polar orbit ($i = 90^\circ$) whose orbital plane is fixed in inertial space (constant RAAN Ω):

- **Declination:** the satellite direction swings from south to north and back, so over one orbit

$$\delta: \quad 0^\circ \rightarrow +90^\circ \rightarrow 0^\circ \rightarrow -90^\circ \rightarrow 0^\circ,$$

at successive equator crossings and pole crossings.

- **Right ascension:** for $i = 90^\circ$ the orbital plane contains Earth's spin axis, so the projection of the satellite position onto the equatorial plane lies along the *line of nodes*. As a result, the satellite's RA stays near the node directions:

$$\alpha \approx \Omega \quad \text{on one half-orbit,} \quad \alpha \approx \Omega + 180^\circ \quad \text{on the opposite half-orbit.}$$

Exactly at the poles ($\delta = \pm 90^\circ$), RA is ill-defined because the projection onto the equatorial plane is zero.

Problem #4

Orbital orientation angles: The three angles Ω (right ascension of the ascending node-RAAN), i (inclination), and ω (argument of perigee) describe how the *orbital plane* and the *ellipse* are oriented relative to Earth's *equatorial plane* and an inertial reference direction (First Point of Aries / vernal equinox direction).

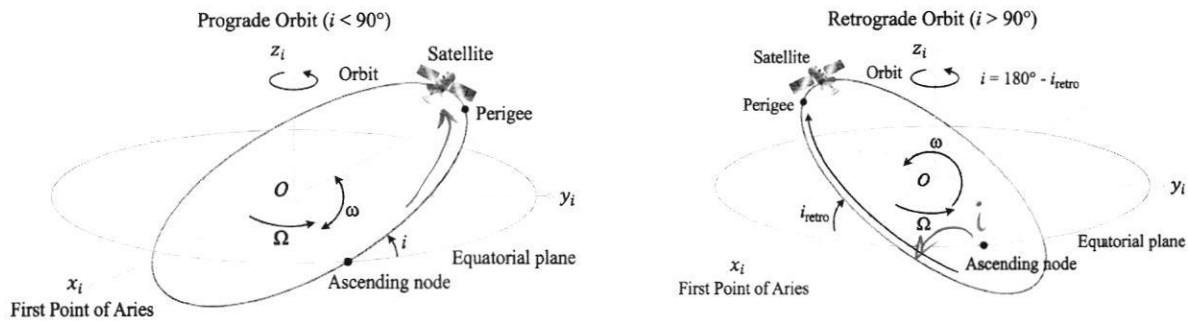


Figure 2: Prograde vs. retrograde geometry for the classical orbital elements Ω (right ascension of the ascending node), i (inclination), and ω (argument of perigee).

The following definitions refer to Fig. 2.

(a) Inclination i .

Inclination i is the angle between the orbital plane and Earth's equatorial plane.

Units: degrees ($^\circ$) or radians.

$$0^\circ \leq i \leq 180^\circ.$$

- **Prograde orbit:** $0^\circ \leq i < 90^\circ$ (motion in the direction of Earth's rotation).
- **Retrograde orbit:** $90^\circ < i \leq 180^\circ$ (motion opposite to Earth's rotation).

Note (retrograde figure): My illustrations show an acute angle (for easier drawing purpose) $i_{\text{retro}} \in [0^\circ, 90^\circ)$ defined by

$$i_{\text{retro}} = 180^\circ - i \quad \Longleftrightarrow \quad i = 180^\circ - i_{\text{retro}},$$

but the *classical* orbital element is $i \in (90^\circ, 180^\circ]$ for retrograde motion.

(b) **Right ascension of the ascending node (RAAN) Ω .**

The **line of nodes** is the intersection of the orbital plane with the equatorial plane. The **ascending node** is the crossing from south to north.

RAAN Ω is measured *in the equatorial plane* from the inertial reference direction (First Point of Aries) to the ascending node direction, measured eastward.

Units: degrees ($^\circ$) or radians.

$$0^\circ \leq \Omega < 360^\circ.$$



(c) **Argument of perigee ω .**

The argument of perigee ω is measured *in the orbital plane* from the ascending node direction to the perigee direction, measured in the direction of motion along the orbit.

Units: degrees ($^\circ$) or radians.

$$0^\circ \leq \omega < 360^\circ.$$

(For retrograde motion, the direction of motion reverses, but ω is still measured from the ascending node to perigee *along the direction of motion*.)



Problem #5

Intelsat 1 is in a geostationary orbit above the equator, so its **subsattellite point** (the point on Earth directly beneath it) is at latitude 0° and longitude l_s . For an Earth station, the **look angles** are:

- γ : **central angle** between the station and the subsattellite point (as seen from Earth's center).
- El : **elevation angle** above the local horizon (negative means below the geometric horizon).
- Az : **azimuth angle**, measured clockwise from true north.

A quick sketch/map is provided below to predict the correct azimuth quadrant (SE/SW/NE/NW) before detailed calculations.

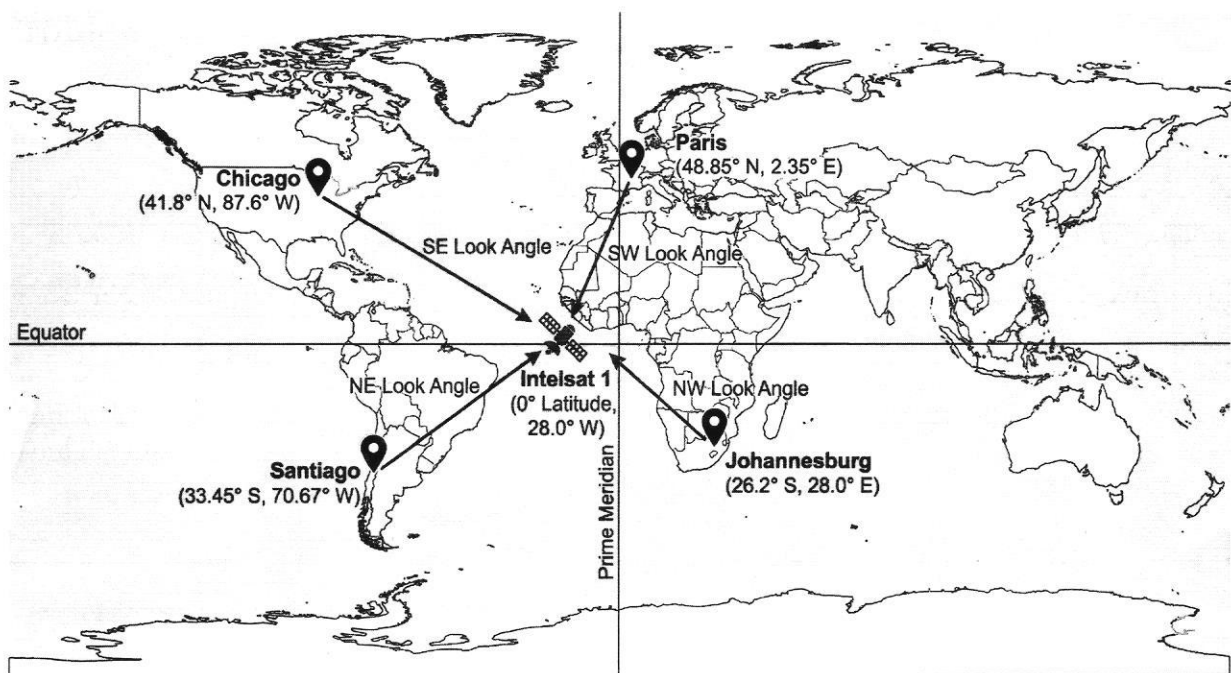


Figure 3: Geographic intuition for look-angle quadrants to Intelsat 1 at $(0^\circ, 28^\circ\text{W})$: Chicago (SE), Paris (SW), Santiago (NE), Johannesburg (NW). ✓

Initial data

- Satellite longitude (also subsattellite-point longitude for GEO):

$$l_s = 28.0^\circ \text{ W} \equiv -28.0^\circ \quad (\text{east } +, \text{ west } -).$$

- Earth radius $R_e = 6378.137$ km, GEO radius $r_s = 42164.17$ km.
- Constant:

$$k \triangleq \frac{r_s}{R_e} = \frac{42164.17}{6378.137} = 6.6107345.$$

All trigonometric functions are evaluated in **degrees**.

(a) Look-angle formulas

Let the Earth station latitude be L_e (north +, south -) and longitude be l_e (east +, west -). Define the longitude separation

$$\Delta l \triangleq l_s - l_e$$

$$\cos \gamma = \cos L_e \cos(\Delta l) \quad (\gamma: \text{central angle station} \leftrightarrow \text{subsatellite point}) \quad (1)$$

$$El = \tan^{-1} \left(\frac{k - \cos \gamma}{\sin \gamma} \right) - \gamma \quad (El: \text{elevation angle above local horizon}) \quad (2)$$

$$\alpha = \tan^{-1} \left(\frac{\tan |\Delta l|}{\sin |L_e|} \right) \quad (\alpha: \text{auxiliary angle used for azimuth}) \quad (3)$$

Visibility check: for GEO, the geometric horizon occurs at

$$\gamma_{\max} = \cos^{-1} \left(\frac{R_e}{r_s} \right) = \cos^{-1} \left(\frac{1}{k} \right) \approx 81.30^\circ.$$

If $\gamma > \gamma_{\max}$, then $El < 0^\circ$ (below the geometric horizon).

Azimuth Az (clockwise from true north) depends on hemisphere and whether the satellite is east/west of the station:

- **Northern Hemisphere** ($L_e > 0$): satellite to SE $\Rightarrow Az = 180^\circ - \alpha$, satellite to SW $\Rightarrow Az = 180^\circ + \alpha$.
- **Southern Hemisphere** ($L_e < 0$): satellite to NE $\Rightarrow Az = \alpha$, satellite to NW $\Rightarrow Az = 360^\circ - \alpha$.

City 1: Chicago, USA ($L_e = 41.8^\circ$, $l_e = -87.6^\circ$)

$$\text{Step 1: } \Delta l = l_s - l_e = (-28.0^\circ) - (-87.6^\circ) = 59.6^\circ. \quad \checkmark$$

$$\begin{aligned} \text{Step 2: } \cos \gamma &= \cos L_e \cos(\Delta l) = \cos(41.8^\circ) \cos(59.6^\circ) = (0.7455)(0.5060) = 0.3772 \\ \Rightarrow \gamma &= \cos^{-1}(0.3772) = 67.84^\circ, \quad \sin \gamma = \sin(67.84^\circ) = 0.9261. \end{aligned}$$

$$\begin{aligned} \text{Step 3: } El &= \tan^{-1} \left(\frac{k - \cos \gamma}{\sin \gamma} \right) - \gamma = \tan^{-1} \left(\frac{6.6107345 - 0.3772}{0.9261} \right) - 67.84^\circ \\ &= \tan^{-1}(6.7308) - 67.84^\circ = 81.55^\circ - 67.84^\circ = 13.71^\circ. \end{aligned}$$

$$\text{Step 4: } \alpha = \tan^{-1} \left(\frac{\tan |\Delta l|}{\sin |L_e|} \right) = \tan^{-1} \left(\frac{\tan(59.6^\circ)}{\sin(41.8^\circ)} \right) = \tan^{-1} \left(\frac{1.7045}{0.6665} \right) = 68.64^\circ.$$

$$\text{Step 5: } Az = 180^\circ - \alpha \quad (\text{NH, satellite east} \Rightarrow \text{SE}) = 180^\circ - 68.64^\circ = 111.36^\circ.$$

$$\boxed{\Delta l = 59.6^\circ, \gamma = 67.84^\circ, El = 13.71^\circ, \alpha = 68.64^\circ, Az = 111.36^\circ \text{ (SE)}}$$

City 2: Santiago, Chile ($L_e = -33.45^\circ$, $l_e = -70.67^\circ$)

Following the same procedure as City 1 using $\Delta l = l_s - l_e$, and the azimuth quadrant rules:

$$\Delta l = (-28.0^\circ) - (-70.67^\circ) = 42.67^\circ.$$

Since $L_e < 0$ (Southern Hemisphere) and $l_s = -28^\circ > -70.67^\circ = l_e$ (satellite is **east** of the station), the satellite appears in the **NE** quadrant, hence $Az = \alpha$.

$$\gamma = 52.16^\circ, \quad El = 30.34^\circ, \quad \alpha = 59.12^\circ, \quad Az = 59.12^\circ \text{ (NE)}$$

City 3: Paris, France ($L_e = 48.85^\circ$, $l_e = +2.35^\circ$)

Applying the same steps:

$$\Delta l = (-28.0^\circ) - (2.35^\circ) = -30.35^\circ.$$

Since $L_e > 0$ (Northern Hemisphere) and $l_s = -28^\circ < +2.35^\circ = l_e$ (satellite is **west** of the station), the satellite appears in the **SW** quadrant, hence $Az = 180^\circ + \alpha$.

$$\gamma = 55.40^\circ, \quad El = 26.84^\circ, \quad \alpha = 37.87^\circ, \quad Az = 217.87^\circ \text{ (SW)}$$

City 4: Johannesburg, South Africa ($L_e = -26.2^\circ$, $l_e = +28.0^\circ$)

Applying the same steps:

$$\Delta l = (-28.0^\circ) - (28.0^\circ) = -56.0^\circ.$$

Since $L_e < 0$ (Southern Hemisphere) and $l_s = -28^\circ < +28^\circ = l_e$ (satellite is **west** of the station), the satellite appears in the **NW** quadrant, hence $Az = 360^\circ - \alpha$.

$$\gamma = 59.88^\circ, \quad El = 22.06^\circ, \quad \alpha = 73.42^\circ, \quad Az = 286.58^\circ \text{ (NW)}$$

Location	(Lat, Lon)	γ	El	α	Case	Az
Chicago	41.8° N, 87.6° W	67.84°	13.71°	68.64°	SE	111.36°
Santiago	33.45° S, 70.67° W	52.16°	30.34°	59.12°	NE	59.12°
Paris	48.85° N, 2.35° E	55.40°	26.84°	37.87°	SW	217.87°
Johannesburg	26.2° S, 28.0° E	59.88°	22.06°	73.42°	NW	286.58°

Table 2: Computed look angles for Intelsat 1 at $l_s = 28^\circ$ W (all angles in degrees).

(b) Suitability for C, Ku, and Ka bands

Using the stated elevation thresholds $El > 5^\circ$ (C), $El > 10^\circ$ (Ku), and $El > 20^\circ$ (Ka):

- **Chicago** ($El = 13.71^\circ$): supports **C** and **Ku**, but **not Ka**.
- **Santiago** ($El = 30.34^\circ$): supports **C**, **Ku**, **Ka**.
- **Paris** ($El = 26.84^\circ$): supports **C**, **Ku**, **Ka**.
- **Johannesburg** ($El = 22.06^\circ$): supports **C**, **Ku**, **Ka**.

(c) Alternative for any unsuitable station

Chicago fails the Ka-band criterion ($El \leq 20^\circ$). A suitable alternative in North America is **Boston, MA** ($L_e = 42.36^\circ$, $l_e = -71.06^\circ$).

$$\begin{aligned}\Delta l &= l_s - l_e = (-28.0^\circ) - (-71.06^\circ) = 43.06^\circ, \\ \cos \gamma &= \cos L_e \cos(\Delta l) = \cos(42.36^\circ) \cos(43.06^\circ) = (0.73858)(0.73130) = 0.53989 \\ \Rightarrow \gamma &= \cos^{-1}(0.53989) = 57.32^\circ, \quad \sin \gamma = \sin(57.32^\circ) = 0.84174, \\ El &= \tan^{-1}\left(\frac{k - \cos \gamma}{\sin \gamma}\right) - \gamma = \tan^{-1}\left(\frac{6.6107345 - 0.53989}{0.84174}\right) - 57.32^\circ \\ &= \tan^{-1}(7.21310) - 57.32^\circ = 82.11^\circ - 57.32^\circ = \mathbf{24.79^\circ}.\end{aligned}$$

Since $24.79^\circ > 20^\circ$, Boston supports **C, Ku, and Ka** under the given thresholds.

